4.4 Indeterminate Forms and L'Hôpital's Rule

Name _____ Period ____ Date _____

AP Calculus 4.4 · Student sheet

I can…

- identify an indeterminate form.
- use L'Hôpital's Rule for the forms 0/0 and ∞/∞.
- rewrite other indeterminate forms so L'Hôpital's Rule applies.

Vocabulary · fill in the missing word

Indeterminate form — a limit like $\frac{0}{0}$ or ∞/∞ whose value can't be found by _____ alone

L'Hôpital's Rule — if the limit of $\frac{f}{g}$ is $\frac{0}{0}$ or ∞/∞, it _____ the limit of $\frac{f'}{g'}$

Other indeterminate forms — _____

Part 1 · Table it

$(e^x - 1)/x$ near $x = 0$.

x	0.1	0.01	0.001
$(e^x - 1)/x$			

Part 2 · Notes and examples (fill in each note, then work the example)

L'Hôpital's Rule: if the limit of f(x)/g(x) is 0/0 or ∞/∞, then it equals the limit of _____.

Example 1

Find the limit as $x \to 0$ of $\frac{\sin(3x)}{2x}$.

Check the form first: use the rule only for 0/0 or _____.

Example 2

Find the limit as $x \to 2$ of $\frac{x^3 - 8}{x - 2}$.

Example 3

Find the limit as $x \to \infty$ of $\frac{x^2}{e^x}$.

Use it again: if the new limit is still 0/0, apply L'Hôpital's Rule _____.

Example 4

Find the limit as $x \to 0$ of $\dfrac{1 - \cos x}{x^2}$.

Forms $1^{\wedge\infty}$, 0^0, ∞^0: take ln y, find its limit L, then the answer is _____.

Example 5

Find the limit as $x \to \infty$ of $(1 + \dfrac{2}{x})^x$.

Part 3 · Partner practice

Partner 1

Find the limit as $x \to 0$ of $\dfrac{e^{4x} - 1}{x}$.

Partner 2

(AP® Practice) The limit as $x \to 0$ of $\dfrac{e^x - 1}{\sin(2x)}$ is

(A) $\dfrac{1}{2}$ (B) 2 (C) 0 (D) does not exist